# Mridul Mathur

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## Education

**SRM Institute of Science and Technology** — Chennai, TN
*Bachelor of Technology in Computer Science and Engineering (AI & ML) – GPA: 8.53* — *Aug. 2024 – May 2028*

**Narayana E-Techno School** — Bengaluru, KA
*CBSE Sr. Secondary Schooling* — *June 2023 – Mar. 2024*

## Experience

**Infinitra Innovations** — Sept. 2025 – Present
*Intern – Full Stack Development (AWS Cloud)*

- Built and deployed production software spanning data pipelines, model inference, REST APIs, and AWS services including Lambda, S3, and CloudWatch.
- Applied software engineering principles to design and implement data-driven applications with database pipeline integration and reliable production workloads.
- Improved application efficiency and API response times using Lambda cold-start mitigation, performance optimization, and CloudWatch monitoring for latency-sensitive workloads.
- Collaborated with product stakeholders on feature requirements, system architecture, implementation, debugging, testing, and technical documentation.

**AWS Student Builder Group @ SRMIST** — June 2026 – Present
*Associate – AI & Machine Learning*

- Lead a student engineering team delivering software and AI applications from ideation through development and deployment, owning technical direction and delivery quality.
- Drive engineering initiatives against defined technical targets, coordinating implementation, testing, debugging, and delivery across student projects.
- Build agentic systems and LLM applications while mentoring AI/ML members and adopting new technologies to solve technical problems.

## Projects

**APX IQ** | *Python, FastAPI, PostgreSQL, TimescaleDB, Socket.IO, FastF1, UDP*

- Engineered a real-time software platform ingesting EA Sports F1 telemetry over UDP at 60 Hz, supporting F1 2020–2025 packet formats through modular adapters.
- Designed independently runnable ingestion, API, and UI services with Socket.IO streaming and PostgreSQL/TimescaleDB persistence for high-frequency telemetry data.
- Developed performance analysis algorithms for distance alignment, corner detection, lap delta, braking, thermal/ERS analysis, battle prediction, and hardware profiling.
- Integrated FastF1 reference data and implemented AI-generated driver debriefs using an Ollama–Gemini–template fallback architecture.
- Implemented software testing and quality gates covering unit/integration tests, linting, type checking, and production builds.

**ARIA** | *Python, FastAPI, TypeScript, Next.js, Neo4j, Qdrant, Docker*

- Architected an autonomous revenue recovery engine analyzing multi-tier payment graphs, achieving 94 root-cause attribution accuracy vs 0.0 baseline across a 62-test chaos battery.
- Implemented a hybrid retrieval pipeline combining Neo4j graph traversal with Qdrant vector embeddings, filtered via cross-encoder re-ranking to isolate root cause failure vectors.
- Engineered deterministic fallback state machines with 0.0 unsafe action rates and 99.88 non-intervention precision, preventing cascading gateway outages.

## Technical Skills

**Languages**: Python, C++, Java, TypeScript, JavaScript, SQL
**Frameworks**: FastAPI, Node.js, Express, React, Next.js
**Computer Science**: Data Structures & Algorithms, Object-Oriented Programming, Algorithms, Software Engineering, Software Design, Problem Solving
**Systems & Data**: PostgreSQL, TimescaleDB, REST APIs, Data Pipelines, Distributed Systems, Real-Time Systems
**Cloud & Engineering**: AWS Lambda, S3, API Gateway, CloudWatch, Docker, CI/CD, Software Testing, Monitoring, Observability, Performance Optimization, Reliability